ELEC70051
EE9-AO6

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE PART IV: MEng and ACGI

RADIO FREQUENCY SUBSYSTEMS

Tuesday, 7 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper. Please follow the instructions on page 1 for information on which questions you should answer.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S. Lucyszyn
Second Marker(s) : W.T. Pike

© Imperial College London

Special instructions for invigilators

This is an OPEN BOOK examination.

Special instructions for students

There are 4 (four) questions in total.

Question 1 is COMPULSARY and carries a weighting of 50%.

Answer 2 (two) out of the remaining 3(three) questions, each carrying a weighting of 25%.

All variables and acronyms have their usual meaning.

1. This question relates to power link budgets.

a)

i) A single-stage subsystem has a power gain G and noise factor F. Using the well-known relationship for the noise factor, in terms of signal-to-noise ratios (assume room-temperature conditions), prove that the output noise power $No = G \cdot F \cdot Ni$.

[2]

ii) For a subsystem with three cascaded stages, give the expression for output noise power in terms of the individual noise power contributions and the individual power gains for the stages. Using this expression, derive the corresponding Friis' Formula for the three-stage subsystem, in terms of the individual noise factors and their power gain.

[6]

iii) If the first stage of multi-stage cascaded subsystem employs an LNA, using the Friis' Formula, justify why this first stage dominates the complete subsystem in terms of noise performance.

[2]

iv) Why are LNAs usually much more expensive than the following stages in a receiver subsystem? Briefly explain why LNAs are not generally found in silicon integrated circuits.

[2]

v) What is the generally accepted room temperature (in Kelvin) used within standard test laboratories? Comment on how this compares to temperatures that have to meet military standards and/or in space environments

[2]

b) Consider the LNB shown in Figure 1.

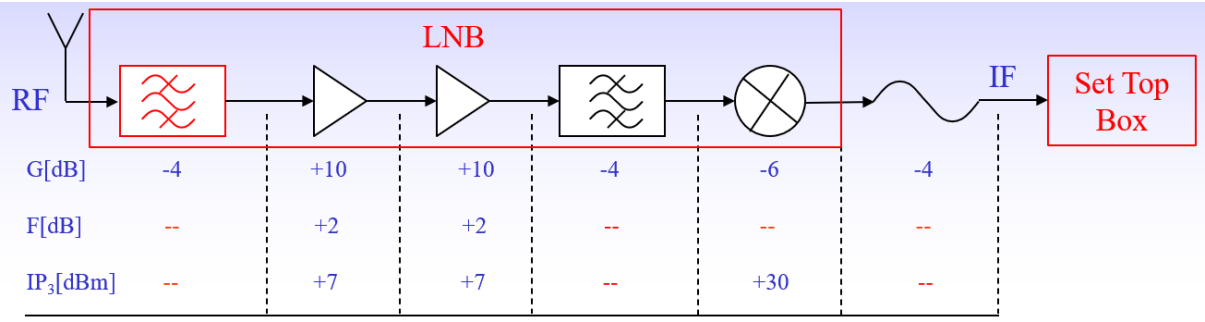


Figure 1.

i) Spell out the abbreviation LNB.

[2]

ii) What domestic application uses an LNB and state the approximate frequency ranges for the RF input and IF output.

[2]

iii) What is the normal role of the first BPF in the LNB chain (in red).

[2]

iv) What is the normal role of the second BPF in the LNB chain.

[2]

v) Given that the mixer has a conversion loss, what does this suggest about its implementation (i.e., is it active or passive?). State your reasoning.

[2]

vi) What is a suitable IP3 value for the passive components? State your reasoning.

[2]

c) From Figure 1, showing all your working out, calculate the following at the output of each stage:

i) Gain (in dB).

[4]

ii) Noise Figure (in dB).

[8]

iii) Third-order Intercept Point (in dBm).

[8]

iv) If the first BPF (in red) is completely removed, comment on what you would expect to see in the overall Gain, Noise Figure and Third-order Intercept Point.

[4]

2. Consider the ideal 5-port network illustrated in Figure 2, consisting of only three couplers and four unspecified square blocks.

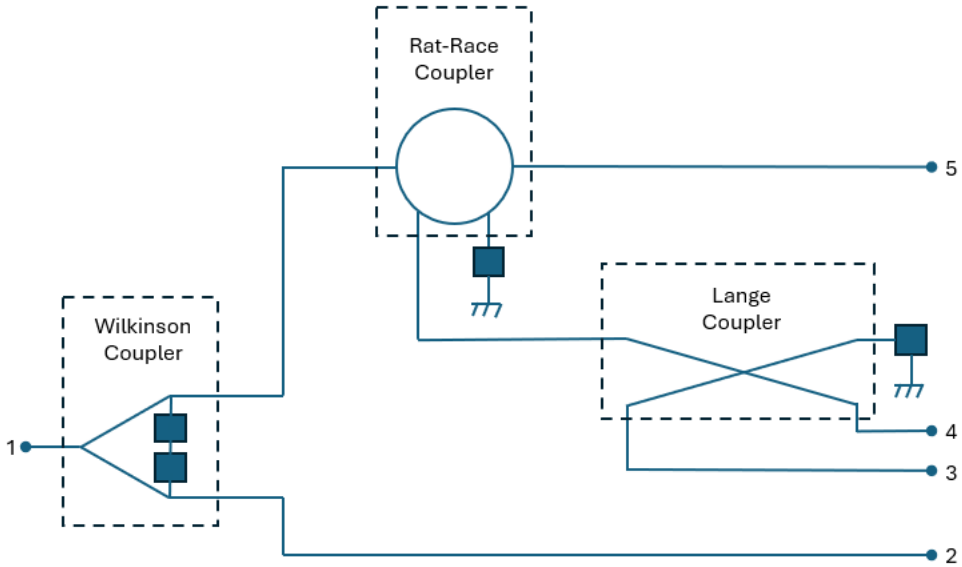


Figure 2.

a) Sketch the individual couplers and clearly indicate each port number. Write down the ideal S-parameter matrices for each of the ideal couplers operating at their centre frequency:

(i) Wilkinson.

[2]

(ii) Rat-Race.

[2]

(iii) Lange.

[2]

b) What do the square (filled-in) boxes represent? With those associated with the Rat-Race and Lange couplers, what are they usually called and what is the ideal S-parameter matrix for these port terminations?

[3]

c) By inspection, determine the ideal 5×5 S-parameter matrix for the complete 5-port network. Hint: start by clearly stating all the simplifying assumptions.

[9]

d) Given the 5-port network shown in Figure 2, four amplifiers and a data-driven single-pole four-throw (SP4T) switch, design a simple QPSK modulator and draw the resulting I-Q constellation. The gain of the individual amplifiers should be chosen so that all the QPSK symbols are located on a unit circle, centred on the origin of the I-Q chart. Hint: start by clearly stating all the simplifying assumptions.

[7]

3. a) Derive, from first principles, the general radar range equation and include all units. Assume the following lossless system:

- The transmitter has an output power P_T feeding an antenna having a power gain G_T that is located at a range R_T from the target.
- The target has a radar cross section σ.
- The receiver has an input power P_R delivered by an antenna having a power gain G_R that is located at a range R_R from the target.

[5]

b) Two amateur radio enthusiasts, living 1 km apart, communicate at 432 MHz using the overhead Moon as a passive satellite (i.e., a reflecting target). The distance to the Moon from both enthusiasts is R_M = 381,500 km and the diameter of the Moon is D_M = 3,500 km. Neglect the effects of the Earth's atmosphere.

i) If identical lossless paraboloidal reflector antennas are used at both locations, calculate the power gain in dBi of the antennas if they have a diameter of 9 m. Note that for an ideal lossless paraboloidal reflector antenna its directivity D_o is given by:

$$D_o = 4\pi \frac{Aperture\ Area}{\lambda^2} \tag{3.1}$$

[3]

ii) Comment on the resulting beam efficiency for this application, when considering the angle that the Moon subtends as seen by an observer on the Earth, if the -3 dB beam width θ is given by:

$$\theta = \sqrt{\frac{4\pi}{D_o}} \tag{3.2}$$

[3]

iii) With P_T = 20 dBW, calculate the power at the receiver. As a first-order approximation, assume that the Moon's radar cross section can be modelled as a perfectly reflecting flat circular disc. Neglect ohmic and diffraction losses.

[3]

iv) With an antenna temperature T_A = 100 K and a receiver noise temperature T_{RX} = 75 K, calculate the system noise temperature T_S, input $(G_R\ /T_S)$ and output (S/N) at a receiver having a final IF bandwidth of 7 kHz.
Boltzmann's constant k = 1.38 x 10^{-23} W/Hz/K.

[8]

v) Calculate the minimum theoretically possible propagation delay time between the two ground stations if both antennas suffer from unwanted side lobes.

[3]

4. A lumped-element *L-C* band stop filter (BSF) has to meet the following specifications:

Lower-pass band -3 dB cut-off frequency:	9.25 GHz
Upper-pass band -3 dB cut-off frequency:	10.75 GHz
Worst-case pass-band insertion loss:	0.1 dB
Band-stop attenuation:	> 110 dB
Stop-band bandwidth:	150 MHz
Source impedance, *Zs*:	25 Ω
Load impedance, Z_L:	50 Ω

a) Calculate the centre frequency, pass-band bandwidth *Bp* and prototype *f/fc* ratio.

[5]

b) From the standard filter curves and tables supplied, determine the normalised component values for the low-pass prototype that meet the required specification. Justify your choice of curve and table.

[5]

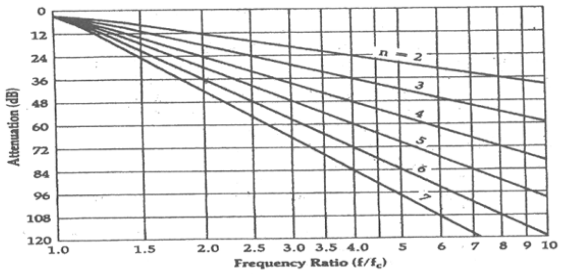
c) Draw the topology for the BPF and calculate the component values.

[10]

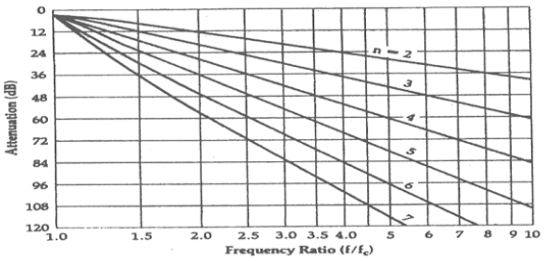
d) Comment on the range of component values and suggest more practical methods for implementing the filter.

[5]

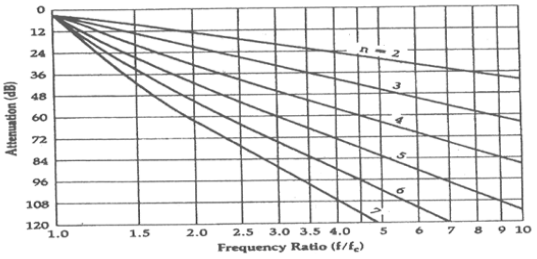
Standard Filter Curves and Tables



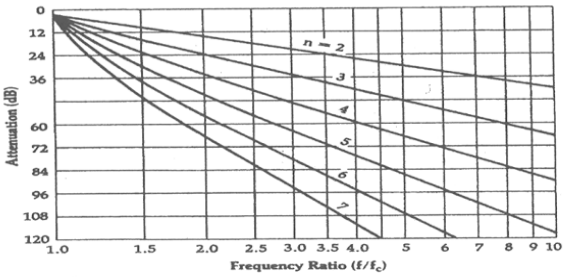
Attenuation characteristics for Butterworth filters.



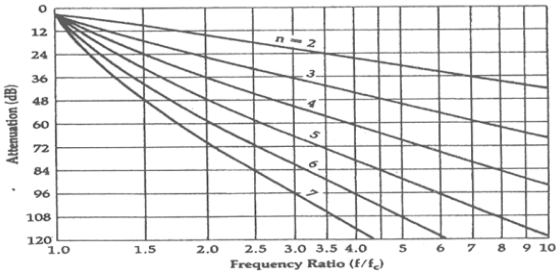
Attenuation characteristics for a Chebyshev filter with 0.01-dB ripple.



Attenuation characteristics for a Chebyshev filter with 0.1-dB ripple.

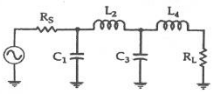


Attenuation characteristics for a Chebyshev filter with 0.5-dB ripple.



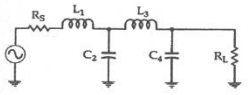
Attenuation characteristics for a Chebyshev filter with 1-dB ripple.

Chebyshev Low-Pass Prototype Element Values for 1.0-dB Ripple

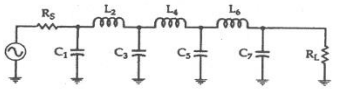


R_S/R_L	C_1	L_2	C_3	L_4
3.000	0.572	3.132		
4.000	0.365	4.600		
8.000	0.157	9.658		
∞	1.213	1.109		
1.000	2.216	1.088	2.216	
0.500	4.431	0.817	2.216	
0.333	6.647	0.726	2.216	
0.250	8.862	0.680	2.216	
0.125	17.725	0.612	2.216	
∞	1.652	1.460	1.108	
3.000	0.653	4.411	0.814	2.535
4.000	0.452	7.083	0.612	2.848
8.000	0.209	17.164	0.428	3.281
∞	1.350	2.010	1.488	1.106

R_L/R_S	L_1	C_2	L_3	C_4

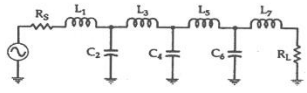


Chebyshev Low-Pass Prototype Element Values for 1.0-dB Ripple

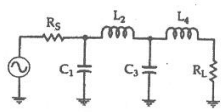


n	R_S/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	2.207	1.128	3.103	1.128	2.207		
	0.500	4.414	0.565	4.653	1.128	2.207		
	0.333	6.622	0.376	6.205	1.128	2.207		
	0.250	8.829	0.282	7.756	1.128	2.207		
	0.125	17.657	0.141	13.961	1.128	2.207		
	∞	1.721	1.645	2.061	1.493	1.103		
6	3.000	0.679	3.873	0.771	4.711	0.969	2.406	
	4.000	0.481	5.644	0.476	7.351	0.849	2.582	
	8.000	0.227	12.310	0.198	16.740	0.726	2.800	
	∞	1.378	2.097	1.690	2.074	1.494	1.102	
7	1.000	2.204	1.131	3.147	1.194	3.147	1.131	2.204
	0.500	4.408	0.566	6.293	0.895	3.147	1.131	2.204
	0.333	6.612	0.377	9.441	0.798	3.147	1.131	2.204
	0.250	8.815	0.283	12.588	0.747	3.147	1.131	2.204
	0.125	17.631	0.141	25.175	0.671	3.147	1.131	2.204
	∞	1.741	1.677	2.155	1.703	2.079	1.494	1.102

n	R_L/R_S	L_1	C_2	L_3	C_4	L_5	C_6	L_7

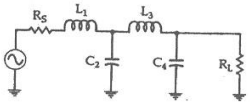


Butterworth Low-Pass Prototype Element Values

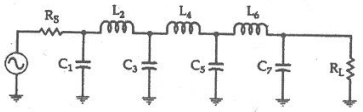


n	R_S/R_L	C_1	L_2	C_3	L_4
2	1.111	1.035	1.835		
	1.250	0.849	2.121		
	1.429	0.697	2.439		
	1.667	0.566	2.828		
	2.000	0.448	3.346		
	2.500	0.342	4.095		
	3.333	0.245	5.313		
	5.000	0.156	7.707		
	10.000	0.074	14.814		
	∞	1.414	0.707		
3	0.900	0.808	1.633	1.599	
	0.800	0.844	1.384	1.926	
	0.700	0.915	1.165	2.277	
	0.600	1.023	0.965	2.702	
	0.500	1.181	0.779	3.261	
	0.400	1.425	0.604	4.064	
	0.300	1.838	0.440	5.363	
	0.200	2.669	0.284	7.910	
	0.100	5.167	0.138	15.455	
	∞	1.500	1.333	0.500	
4	1.111	0.466	1.592	1.744	1.469
	1.250	0.388	1.695	1.511	1.811
	1.429	0.325	1.862	1.291	2.175
	1.667	0.269	2.103	1.082	2.613
	2.000	0.218	2.452	0.883	3.187
	2.500	0.169	2.986	0.691	4.009
	3.333	0.124	3.883	0.507	5.338
	5.000	0.080	5.684	0.331	7.940
	10.000	0.039	11.064	0.162	15.642
	∞	1.531	1.577	1.082	0.383

n	R_L/R_S	L_1	C_2	L_3	C_4

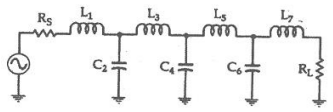


Butterworth Low-Pass Prototype Element Values

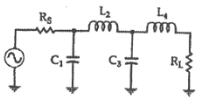


n	R_S/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	0.900	0.442	1.027	1.910	1.756	1.389		
	0.800	0.470	0.866	2.061	1.544	1.738		
	0.700	0.517	0.731	2.285	1.333	2.108		
	0.600	0.586	0.609	2.600	1.126	2.552		
	0.500	0.686	0.496	3.051	0.924	3.133		
	0.400	0.838	0.388	3.736	0.727	3.965		
	0.300	1.094	0.285	4.884	0.537	5.307		
	0.200	1.608	0.186	7.185	0.352	7.935		
	0.100	3.512	0.091	14.095	0.173	15.710		
	∞	1.545	1.694	1.382	0.894	0.309		
6	1.111	0.289	1.040	1.322	2.054	1.744	1.335	
	1.250	0.245	1.116	1.126	2.239	1.550	1.688	
	1.429	0.207	1.236	0.957	2.499	1.346	2.062	
	1.667	0.173	1.407	0.801	2.858	1.143	2.509	
	2.000	0.141	1.653	0.654	3.369	0.942	3.094	
	2.500	0.111	2.028	0.514	4.141	0.745	3.931	
	3.333	0.082	2.656	0.379	5.433	0.552	5.280	
	5.000	0.054	3.917	0.248	8.020	0.363	7.922	
	10.000	0.026	7.705	0.122	15.786	0.179	15.738	
	∞	1.553	1.759	1.553	1.202	0.758	0.259	
7	0.900	0.299	0.711	1.404	1.489	2.125	1.727	1.296
	0.800	0.322	0.606	1.517	1.278	2.334	1.546	1.652
	0.700	0.357	0.515	1.688	1.091	2.618	1.350	2.028
	0.600	0.408	0.432	1.928	0.917	3.005	1.150	2.477
	0.500	0.480	0.354	2.273	0.751	3.553	0.951	3.064
	0.400	0.590	0.278	2.795	0.592	4.380	0.754	3.904
	0.300	0.775	0.206	3.671	0.437	5.761	0.560	5.258
	0.200	1.145	0.135	5.427	0.287	8.526	0.369	7.908
	0.100	2.257	0.067	10.700	0.142	16.822	0.182	15.748
	∞	1.558	1.799	1.659	1.397	1.055	0.656	0.223

n	R_L/R_S	L_1	C_2	L_3	C_4	L_5	C_6	L_7

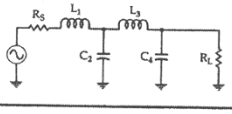


Chebyshev Low-Pass Element Values for 0.01-dB Ripple

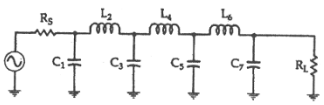


n	R_S/R_L	C_1	L_2	C_3	L_4
2	1.101	1.347	1.483		
	1.111	1.247	1.595		
	1.250	0.943	1.997		
	1.429	0.759	2.344		
	1.667	0.609	2.750		
	2.000	0.479	3.277		
	2.500	0.363	4.033		
	3.333	0.259	5.255		
	5.000	0.164	7.650		
	10.000	0.078	14.749		
	∞	1.412	0.742		
3	1.000	1.181	1.821	1.181	
	0.900	1.092	1.660	1.480	
	0.800	1.097	1.443	1.806	
	0.700	1.160	1.228	2.165	
	0.600	1.274	1.024	2.598	
	0.500	1.452	0.829	3.164	
	0.400	1.734	0.645	3.974	
	0.300	2.216	0.470	5.280	
	0.200	3.193	0.305	7.834	
	0.100	6.141	0.148	15.390	
	∞	1.501	1.433	0.591	
4	1.100	0.950	1.938	1.761	1.046
	1.111	0.854	1.946	1.744	1.165
	1.250	0.618	2.075	1.542	1.617
	1.429	0.495	2.279	1.334	2.008
	1.667	0.398	2.571	1.128	2.461
	2.000	0.316	2.994	0.926	3.045
	2.500	0.242	3.641	0.729	3.875
	3.333	0.174	4.727	0.538	5.209
	5.000	0.112	6.910	0.352	7.813
	10.000	0.054	13.469	0.173	15.510
	∞	1.529	1.694	1.312	0.523

n	R_L/R_S	L_1	C_2	L_3	C_4

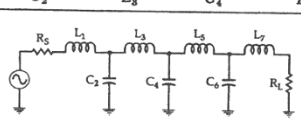


Chebyshev Low-Pass Element Values for 0.01-dB Ripple

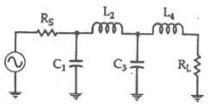


n	R_S/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	0.977	1.685	2.037	1.685	0.977		
	0.900	0.880	1.456	2.174	1.641	1.274		
	0.800	0.877	1.235	2.379	1.499	1.607		
	0.700	0.926	1.040	2.658	1.323	1.977		
	0.600	1.019	0.863	3.041	1.135	2.424		
	0.500	1.166	0.699	3.584	0.942	3.009		
	0.400	1.398	0.544	4.403	0.749	3.845		
	0.300	1.797	0.398	5.772	0.557	5.193		
	0.200	2.604	0.259	8.514	0.368	7.826		
	0.100	5.041	0.127	16.741	0.182	15.613		
	∞	1.547	1.795	1.645	1.237	0.488		
6	1.101	0.851	1.796	1.841	2.027	1.631	0.937	
	1.111	0.760	1.782	1.775	2.094	1.638	1.053	
	1.250	0.545	1.864	1.489	2.403	1.507	1.504	
	1.429	0.436	2.038	1.266	2.735	1.332	1.899	
	1.667	0.351	2.298	1.061	3.167	1.145	2.357	
	2.000	0.279	2.678	0.867	3.768	0.954	2.948	
	2.500	0.214	3.261	0.682	4.667	0.761	3.790	
	3.333	0.155	4.245	0.503	6.163	0.568	5.143	
	5.000	0.100	6.223	0.330	9.151	0.376	7.785	
	10.000	0.048	12.171	0.162	18.105	0.187	15.595	
	∞	1.551	1.847	1.790	1.598	1.190	0.469	
7	1.000	0.913	1.595	2.002	1.870	2.002	1.595	0.913
	0.900	0.816	1.362	2.089	1.722	2.202	1.581	1.206
	0.800	0.811	1.150	2.262	1.525	2.465	1.464	1.538
	0.700	0.857	0.967	2.516	1.323	2.802	1.307	1.910
	0.600	0.943	0.803	2.872	1.124	3.250	1.131	2.359
	0.500	1.080	0.650	3.382	0.928	3.875	0.947	2.948
	0.400	1.297	0.507	4.156	0.735	4.812	0.758	3.790
	0.300	1.669	0.372	5.454	0.546	6.370	0.568	5.148
	0.200	2.242	0.242	8.057	0.360	9.484	0.378	7.802
	0.100	4.701	0.119	15.872	0.178	18.818	0.188	15.652
	∞	1.559	1.867	1.866	1.765	1.563	1.161	0.456

n	R_L/R_S	L_1	C_2	L_3	C_4	L_5	C_6	L_7

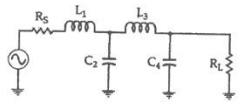


Chebyshev Low-Pass Prototype Element Values for 0.1-dB Ripple

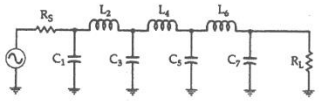


n	R_S/R_L	C_1	L_2	C_3	L_4
2	1.355	1.209	1.638		
	1.429	0.977	1.982		
	1.667	0.733	2.489		
	2.000	0.560	3.054		
	2.500	0.417	3.827		
	3.333	0.293	5.050		
	5.000	0.184	7.426		
	10.000	0.087	14.433		
	∞	1.391	0.819		
3	1.000	1.433	1.594	1.433	
	0.900	1.426	1.494	1.622	
	0.800	1.451	1.356	1.871	
	0.700	1.521	1.193	2.190	
	0.600	1.648	1.017	2.603	
	0.500	1.853	0.838	3.159	
	0.400	2.186	0.660	3.968	
	0.300	2.763	0.486	5.279	
	0.200	3.942	0.317	7.850	
	0.100	7.512	0.155	15.466	
	∞	1.513	1.510	0.716	
4	1.355	0.992	2.148	1.585	1.341
	1.429	0.779	2.348	1.429	1.700
	1.667	0.576	2.730	1.185	2.243
	2.000	0.440	3.227	0.967	2.856
	2.500	0.329	3.961	0.760	3.698
	3.333	0.233	5.178	0.560	5.030
	5.000	0.148	7.607	0.367	7.614
	10.000	0.070	14.887	0.180	15.230
	∞	1.511	1.768	1.455	0.673

n	R_L/R_S	L_1	C_2	L_3	C_4

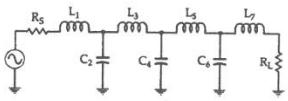


Chebyshev Low-Pass Prototype Element Values for 0.1-dB Ripple

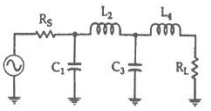


n	R_S/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	1.301	1.556	2.241	1.556	1.301		
	0.900	1.285	1.433	2.380	1.488	1.488		
	0.800	1.300	1.282	2.582	1.382	1.738		
	0.700	1.358	1.117	2.868	1.244	2.062		
	0.600	1.470	0.947	3.269	1.085	2.484		
	0.500	1.654	0.778	3.845	0.913	3.055		
	0.400	1.954	0.612	4.720	0.733	3.886		
	0.300	2.477	0.451	6.196	0.550	5.237		
	0.200	3.546	0.295	9.127	0.366	7.889		
	0.100	6.787	0.115	17.957	0.182	15.745		
	∞	1.561	1.807	1.766	1.417	0.651		
6	1.355	0.942	2.080	1.659	2.247	1.534	1.277	
	1.429	0.735	2.249	1.454	2.544	1.405	1.629	
	1.667	0.542	2.600	1.183	3.064	1.185	2.174	
	2.000	0.414	3.068	0.958	3.712	0.979	2.794	
	2.500	0.310	3.765	0.749	4.651	0.778	3.645	
	3.333	0.220	4.927	0.551	6.195	0.580	4.996	
	5.000	0.139	7.250	0.361	9.261	0.384	7.618	
	10.000	0.067	14.220	0.178	18.427	0.190	15.350	
	∞	1.534	1.884	1.831	1.749	1.394	0.638	
7	1.000	1.262	1.520	2.239	1.680	2.239	1.520	1.262
	0.900	1.242	1.395	2.361	1.578	2.397	1.459	1.447
	0.800	1.255	1.245	2.548	1.443	2.624	1.362	1.697
	0.700	1.310	1.083	2.819	1.283	2.942	1.233	2.021
	0.600	1.417	0.917	3.205	1.209	3.384	1.081	2.444
	0.500	1.595	0.753	3.764	0.928	4.015	0.914	3.018
	0.400	1.885	0.593	4.618	0.742	4.970	0.738	3.855
	0.300	2.392	0.437	6.054	0.556	6.569	0.557	5.217
	0.200	3.428	0.286	8.937	0.369	9.770	0.372	7.890
	0.100	6.570	0.141	17.603	0.184	19.376	0.186	15.813
	∞	1.575	1.858	1.921	1.827	1.734	1.379	0.631

n	R_L/R_S	L_1	C_2	L_3	C_4	L_5	C_6	L_7

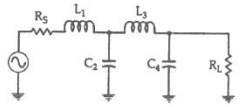


Chebyshev Low-Pass Prototype Element Values for 0.5-dB Ripple

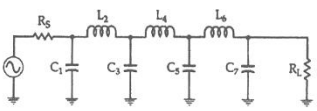


n	R_S/R_L	C_1	L_2	C_3	L_4
2	1.984	0.983	1.950		
	2.000	0.909	2.103		
	2.500	0.564	3.165		
	3.333	0.375	4.411		
	5.000	0.228	6.700		
	10.000	0.105	13.322		
	∞	1.307	0.975		
3	1.000	1.864	1.280	1.834	
	0.900	1.918	1.209	2.026	
	0.800	1.997	1.120	2.237	
	0.700	2.114	1.015	2.517	
	0.500	2.557	0.759	3.436	
	0.400	2.985	0.615	4.242	
	0.300	3.729	0.463	5.576	
	0.200	5.254	0.309	8.225	
	0.100	9.890	0.153	16.118	
	∞	1.572	1.518	0.932	
4	1.984	0.920	2.586	1.304	1.826
	2.000	0.845	2.720	1.238	1.985
	2.500	0.516	3.766	0.869	3.121
	3.333	0.344	5.120	0.621	4.480
	5.000	0.210	7.708	0.400	6.987
	10.000	0.098	15.352	0.194	14.262
	∞	1.436	1.889	1.521	0.913

n	R_L/R_S	L_1	C_2	L_3	C_4



Chebyshev Low-Pass Prototype Element Values for 0.5-dB Ripple



n	R_S/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	1.807	1.303	2.691	1.303	1.807		
	0.900	1.854	1.222	2.849	1.238	1.970		
	0.800	1.926	1.126	3.060	1.157	2.185		
	0.700	2.035	1.015	3.353	1.058	2.470		
	0.600	2.200	0.890	3.765	0.942	2.861		
	0.500	2.457	0.754	4.367	0.810	3.414		
	0.400	2.870	0.609	5.296	0.664	4.245		
	0.300	3.588	0.459	6.871	0.508	5.625		
	0.200	5.064	0.306	10.054	0.343	8.367		
	0.100	9.556	0.153	19.647	0.173	16.574		
	∞	1.630	1.740	1.922	1.514	0.903		
6	1.984	0.905	2.577	1.368	2.713	1.299	1.796	
	2.000	0.830	2.704	1.291	2.872	1.237	1.956	
	2.500	0.506	3.722	0.890	4.109	0.881	3.103	
	3.333	0.337	5.055	0.632	5.699	0.635	4.481	
	5.000	0.206	7.615	0.406	8.732	0.412	7.031	
	10.000	0.096	15.186	0.197	17.681	0.202	14.433	
7	1.000	1.790	1.296	2.718	1.385	2.718	1.296	1.790
	0.900	1.835	1.215	2.869	1.308	2.883	1.234	1.953
	0.800	1.905	1.118	3.076	1.215	3.107	1.155	2.168
	0.700	2.011	1.007	3.364	1.105	3.416	1.058	2.455
	0.600	2.174	0.882	3.772	0.979	3.852	0.944	2.848
	0.500	2.428	0.747	4.370	0.838	2.289	0.814	3.405
	0.400	2.835	0.604	5.295	0.685	5.470	0.669	4.243
	0.300	3.546	0.455	6.867	0.522	7.134	0.513	5.635
	0.200	5.007	0.303	10.049	0.352	10.496	0.348	8.404
	0.100	9.456	0.151	19.649	0.178	20.631	0.176	16.665
	∞	1.646	1.777	2.031	1.789	1.924	1.503	0.895

n	R_L/R_S	L_1	C_2	L_3	C_4	L_5	C_6	L_7

